AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 31 · STUDENT LESSON

Subgroup Performance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 34: calculate subgroup rates before combining them.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Calculate subgroup results with sample sizes. Try the recall before opening notes.

Your learning target

I can calculate **subgroup** results with sample sizes.

I can distinguish missing **evidence** from equal performance.

I can report differences without assigning causes from a table.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

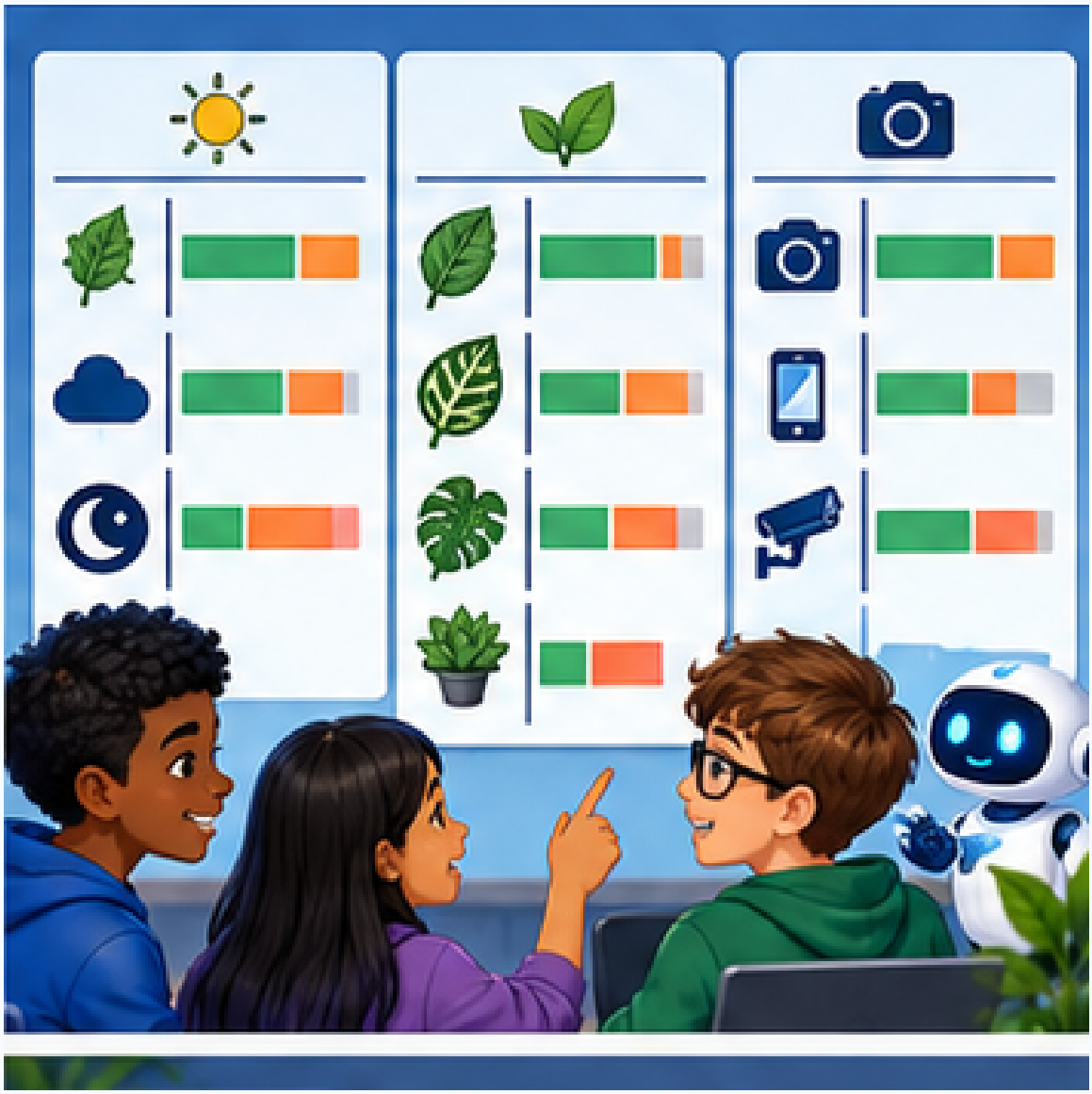
Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Overall metrics can conceal uneven behavior across meaningful conditions or user groups.



Your target

LEARNING TARGET Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

Builder name and experiment date:

Readiness for the connected explanation

What can a pooled error rate hide?

This week's end goal

By the end, I can explain **subgroup** performance using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

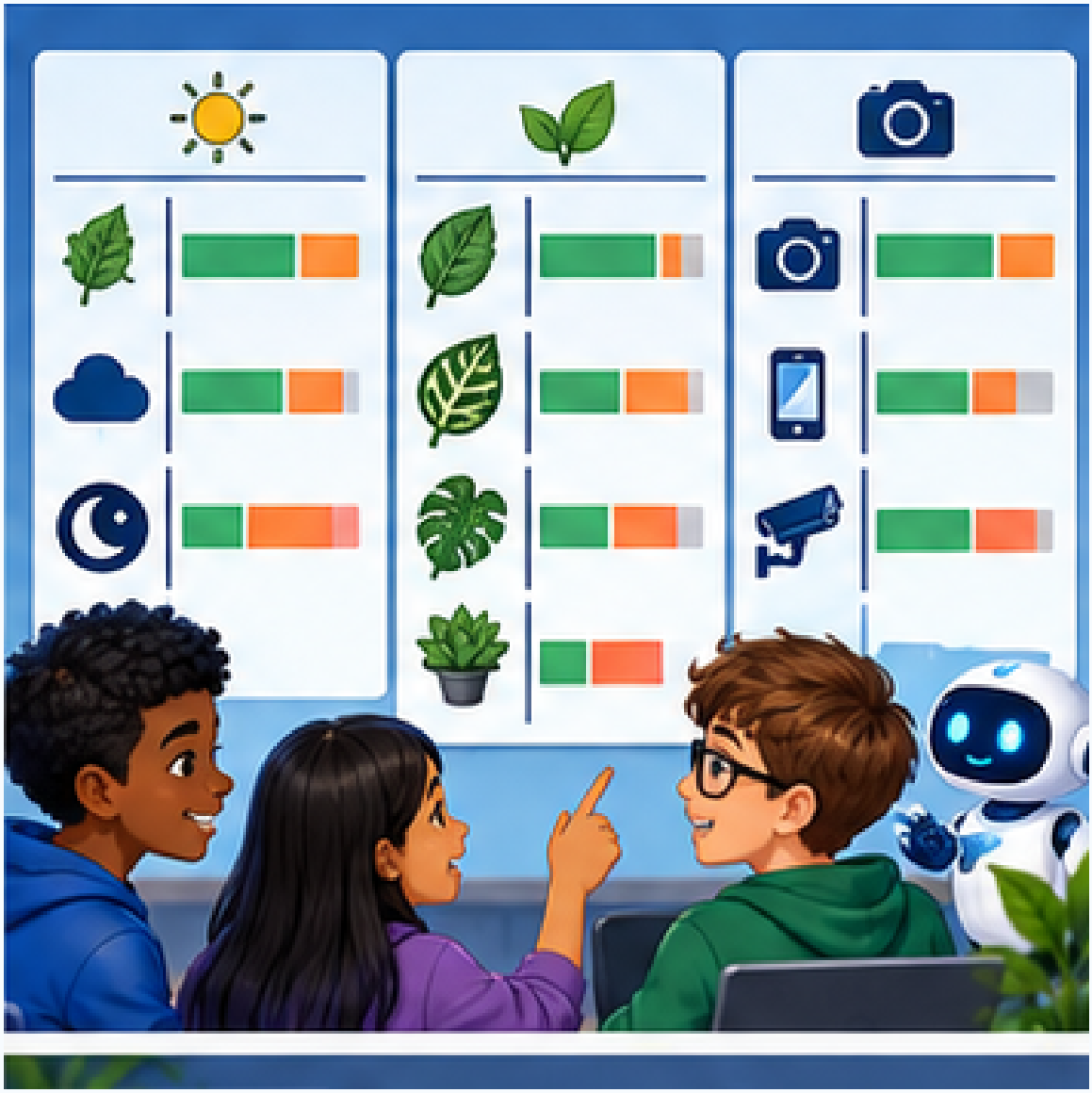
Notice the evidence

> Investigate the supplied Subgroup Performance case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **subgroup** performance to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A single overall score can hide different experiences. Define relevant groups before evaluation and report counts as well as rates. This classroom exercise uses fictional sensor conditions, not personal identities.

Real evaluations involving people require appropriate permissions, privacy and contextual expertise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **subgroup** evaluation.

STUDENT LESSON · 3 OF 18

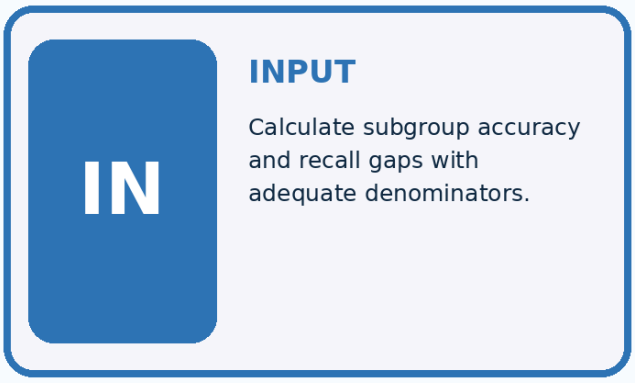
See how it works

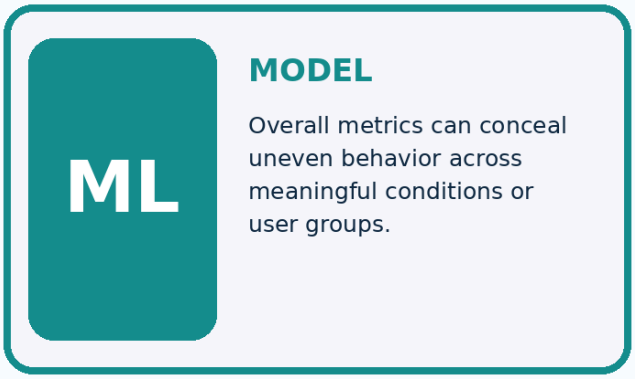
> Follow the mechanism for Subgroup Performance. Connect each labelled input, deciding step and result.

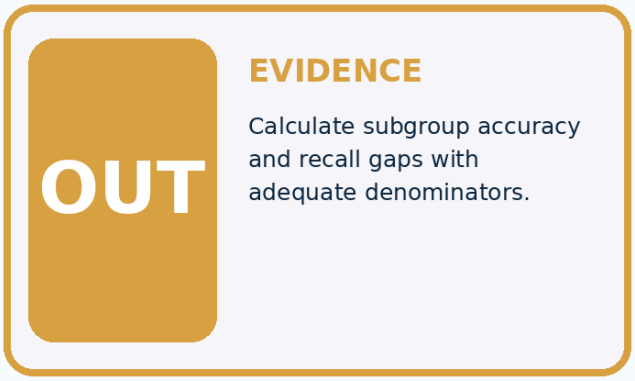
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Calculate **subgroup** accuracy and recall gaps with adequate **denominator**s.

MODEL IDEA Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Daylight: 18/20 correct = 90%; evening: 10/20 = 50%. Accuracy gap = 90% – 50% = 40 percentage points.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Sensor group A has 2 errors in 20 examples and group B has 3 in 5. Error rates are 10 and 60 percent; pooled error is 5/25=20 percent. The group difference is 50 percentage points.

B has only five examples, so its rate is sensitive to each individual result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

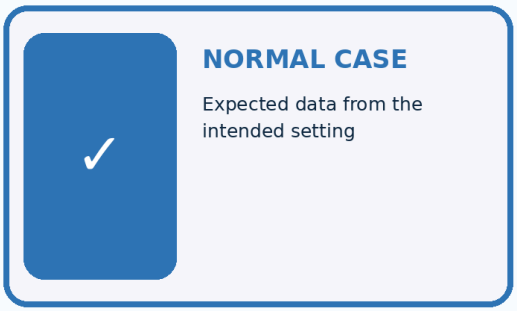
Compare the cases

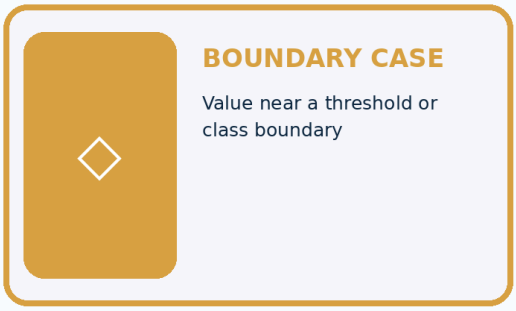
> Compare the supplied cases for Subgroup Performance. Identify the detail that changes the conclusion.

EVALUATION DESIGN

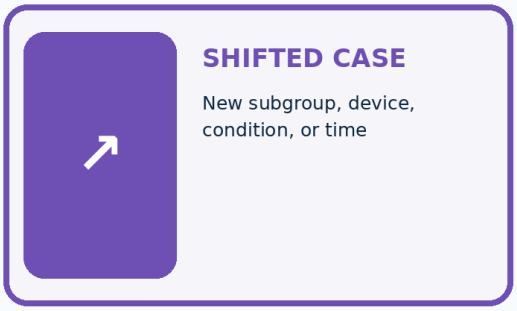
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, **subgroup**, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Shade gains5 new cases with no errors.

Predicted result

Prediction: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.

Why the result follows

Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

STUDENT LESSON · 5 OF 18

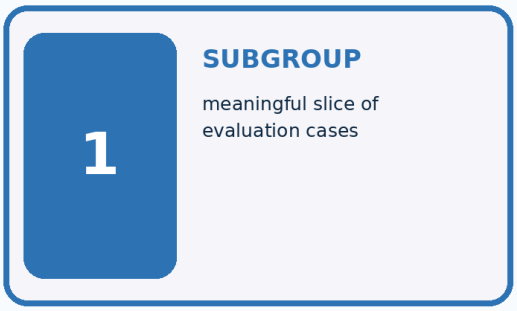
Words that unlock the idea

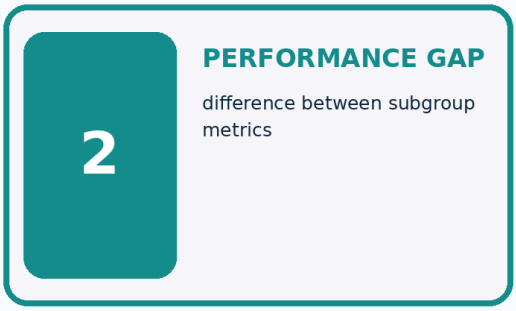
> Use the vocabulary of Subgroup Performance to explain a concrete decision, rather than only memorising definitions.

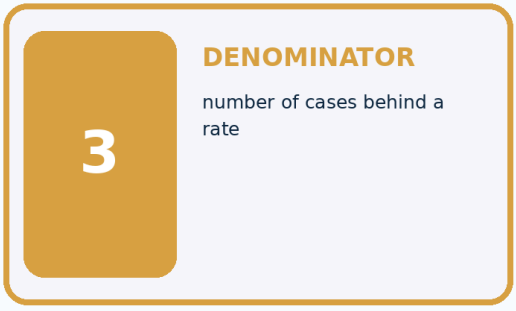
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
subgroup	meaningful slice of evaluation cases	
performance gap	difference between subgroup metrics	
denominator	number of cases behind a rate	
intersection	cases belonging to multiple slices	

MISCONCEPTION TO REPAIR A good overall result guarantees acceptable performance for every **subgroup**.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Subgroup	A defined subset of the evaluated examples
Disaggregation	Reporting results separately for defined groups
Coverage	Which groups and conditions are represented
Disparity	An observed difference between group results

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Subgroup Performance, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 31 • WORKED EVIDENCE

1. Daylight: 18/20 correct = 90%; evening: 10/20 = 50%.
2. Accuracy gap = 90% − 50% = 40 percentage points.
3. Evening spotted recall is 4/10 = 40%, versus daylight 8/10 = 80%.
4. Inspect representation and image quality; report counts and uncertainty before repairing.

Step 1

STEP 1 Daylight: 18/20 correct = 90%; evening: 10/20 = 50%.

Step 2

STEP 2 Accuracy gap = 90% – 50% = 40 percentage points.

Step 3

STEP 3 Evening spotted recall is 4/10 = 40%, versus daylight 8/10 = **80%**.

Step 4

STEP 4 Inspect representation and image quality; report counts and uncertainty before repairing.

CHECK Verify the **denominator**, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Subgroup evaluation at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Group A	2 errors of 20	10 percent
Group B	3 errors of 5	60 percent
Pooled	5 errors of 25	20 percent

Changing one B error to a correct result changes its rate from 60 to 40 percent. Small samples can be unstable without making the observed errors unimportant. A missing group has no measured rate, not zero error. Intersections of conditions may be even smaller.

How does correcting one B error affect its rate? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Subgroup Performance

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

The facts we will use

Supplied facts

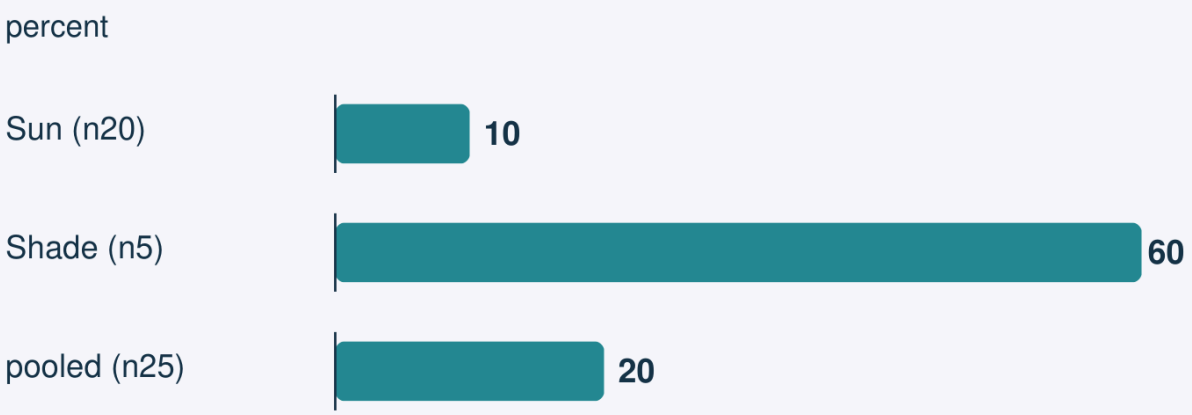
Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/31

Subgroup error rates



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/31

Subgroup Performance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Sun rate

Sun error=2/20=10%.



2. Shade rate

Shade error=3/5=60%.



3. Pooled rate

Total error=5/25=20%, which hides the 50-percentage-point subgroup difference.

CHANGE ONE THING

Shade gains5 new cases with no errors.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Sun rate	Sun error=2/20=10%.
2. Shade rate	Shade error=3/5=60%.
3. Pooled rate	Total error=5/25=20%, which hides the 50-percentage-point subgroup difference.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Small **subgroup** samples have high variability. An observed gap does not by itself establish

discrimination or a causal explanation.

Keep the stated limit

Read the labelled picture: Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Shade gains5 new cases with no errors. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Subgroup Performance claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A good overall result guarantees acceptable performance for every subgroup. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A good overall result guarantees acceptable performance for every **subgroup**.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Equal error rates alone do not establish fairness, and unequal rates alone do not identify discrimination or a cause. Groups may differ in base rates, label quality or conditions.

Use results to ask better questions and plan review rather than label a group as inherently difficult.

Why does equal accuracy not settle whether use is fair?

STUDENT LESSON · 8 OF 18

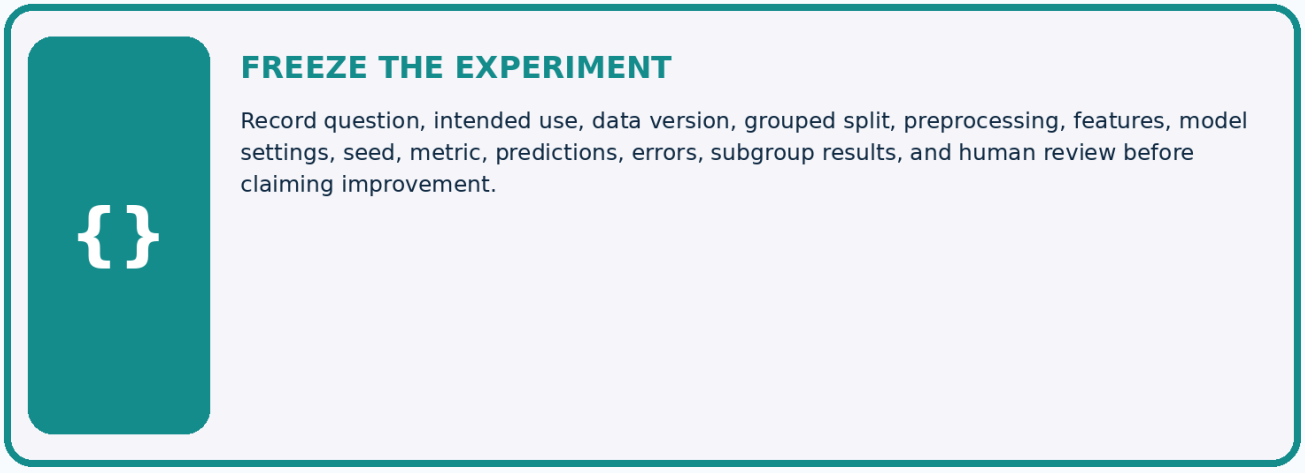
Practise together

Practise Subgroup Performance with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + **denominator**s

PREDICTIONS + errors + **subgroup** slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Choose group definitions for a reason connected to use and possible harm. Retain error types, reference-label quality and sampling method. Avoid searching many groupings merely to find a dramatic or reassuring number.

Report exploratory checks as exploratory.

Practise with less help: Check sensitivity

Use workbook W2 for the connected example "**Subgroup** evaluation". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Subgroup** evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Subgroup Performance. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Slice Audit: compute metrics and denominators for lighting, species, and camera; flag small samples and prioritize one gap.

1. Write the hypothesis and expected **evidence** before running.

2. Slice Audit: compute metrics and **denominator**s for lighting, species, and camera; flag small samples and prioritize one gap.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

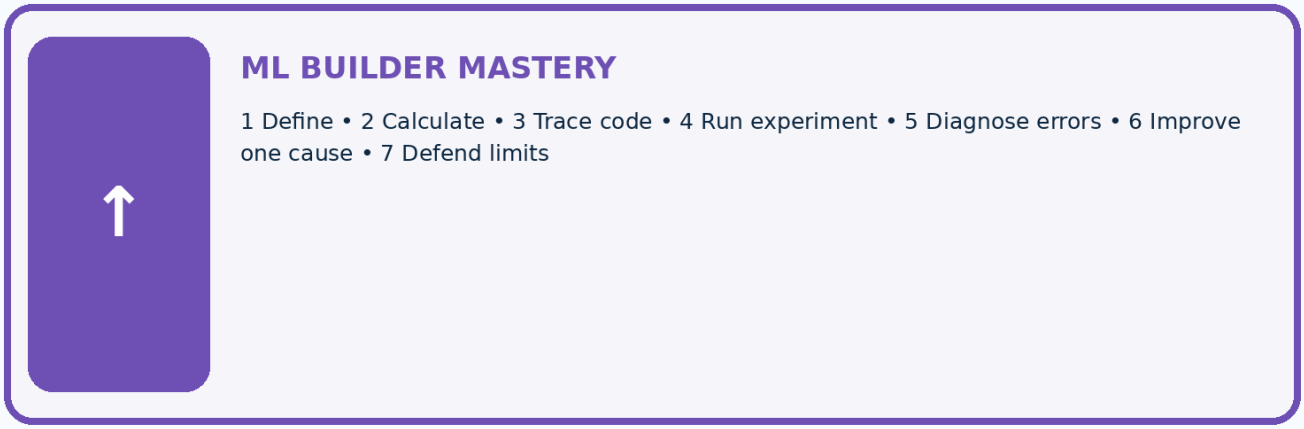
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Subgroup Performance. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **subgroup**, **performance gap**, **denominator**, **intersection**.

CALCULATE Reproduce the worked **evidence** with units and **denominator**.

TRACE Explain the data, code or decision path.

DIAGNOSE A good overall result guarantees acceptable performance for every **subgroup**.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Add subgroup tables and a minimum acceptable threshold to the Plant Classifier release criteria.

THIS WEEK'S CHECKPOINT Add **subgroup** tables and a minimum acceptable threshold to the Plant Classifier release criteria.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

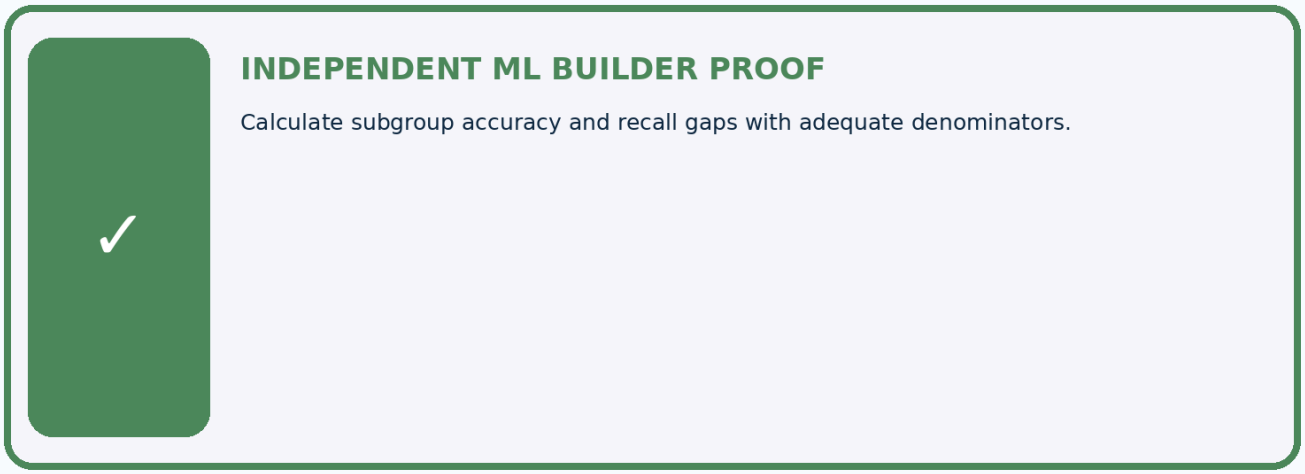
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Calculate **subgroup** accuracy and recall gaps with adequate **denominator**s.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Proxy Features and Fairness.

Study a complete case

> Revisit the complete Subgroup Performance case for supported practice. Repeated examples do not become fresh assessments.

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A has18/20 correct; B has6/10.

The question

Compute **subgroup** results and **evidence** limits.

Worked explanation

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Report subgroup denominators with rates

Evidence or stage	Value or result
Group A	18/20
Group B	6/10

A has18/20 correct; B has6/10.

Compute **subgroup** results and **evidence** limits.

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Add **subgroup** tables and a minimum acceptable threshold to the Plant Classifier release criteria.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Subgroup Performance, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

By the end, I can	How I will show it
Calculate subgroup results with sample sizes	Explain the deciding step, show a trace or test, and state one limit.
Distinguish missing evidence from equal performance	Explain the deciding step, show a trace or test, and state one limit.
Report differences without assigning causes from a table	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Subgroup Performance without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate **subgroup** results with sample sizes

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.

1. Pooled rate

2. Sun rate

3. Shade rate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Pooled rate Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Read the picture in order.

Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Shade gains5 new cases with no errors. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Repair a missing building block

Review Improve Generalization (ai-4/30). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: What fresh **evidence** is needed after adding crossed **training** data?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

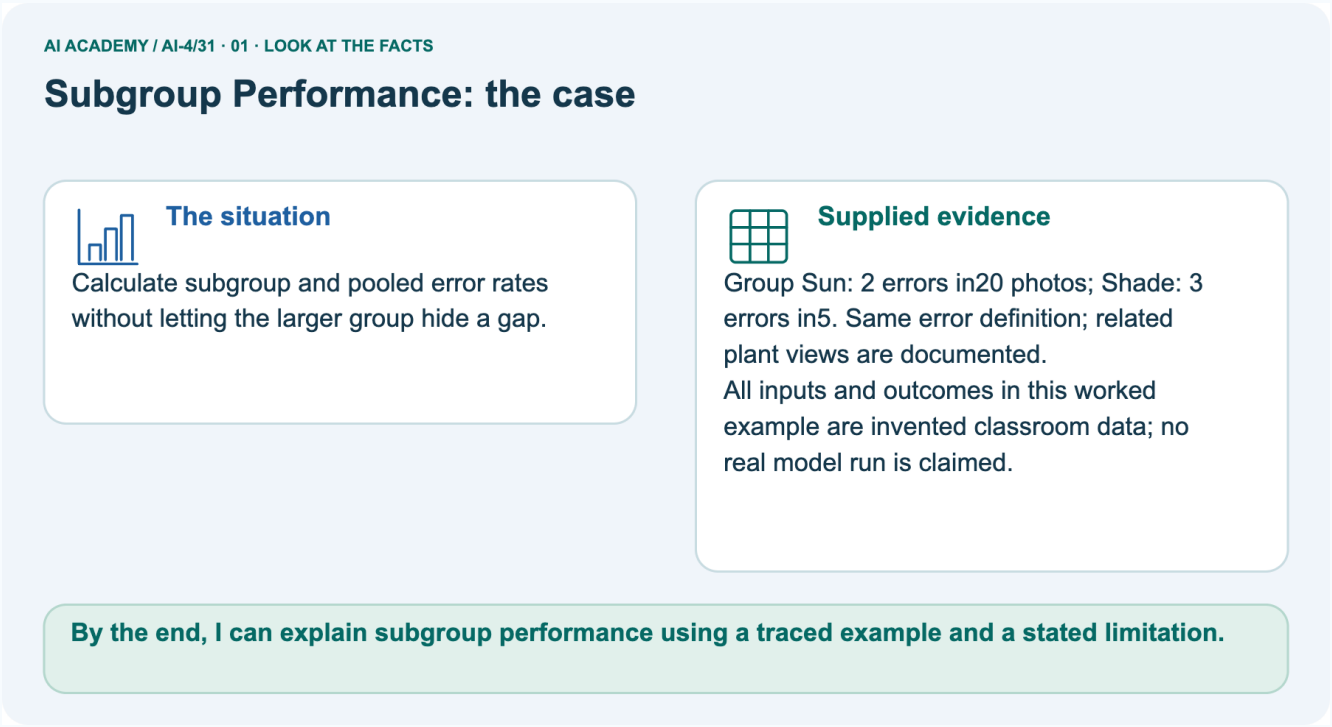
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Subgroup Performance graphic using its labels, deciding evidence and relevant limitation.

01. Subgroup Performance: the case



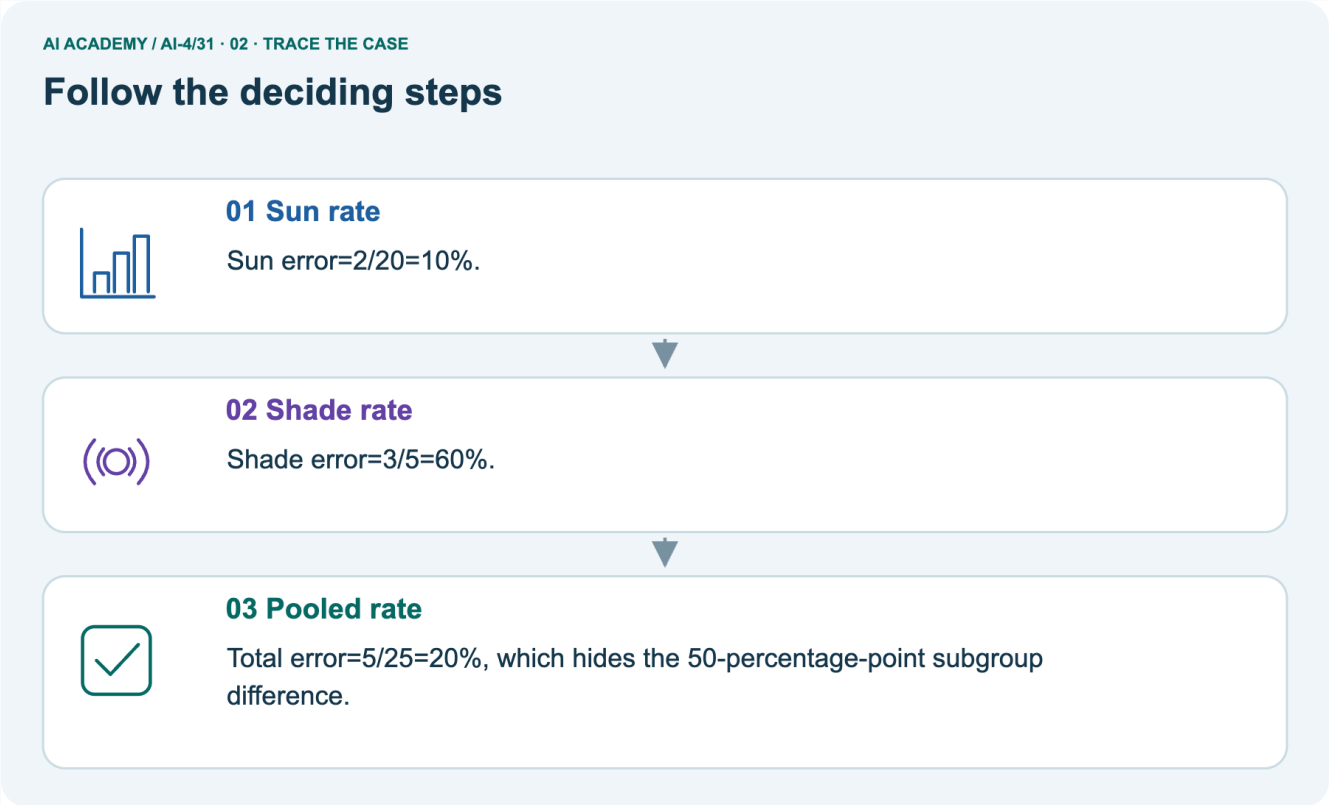
Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap. Supplied facts: Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

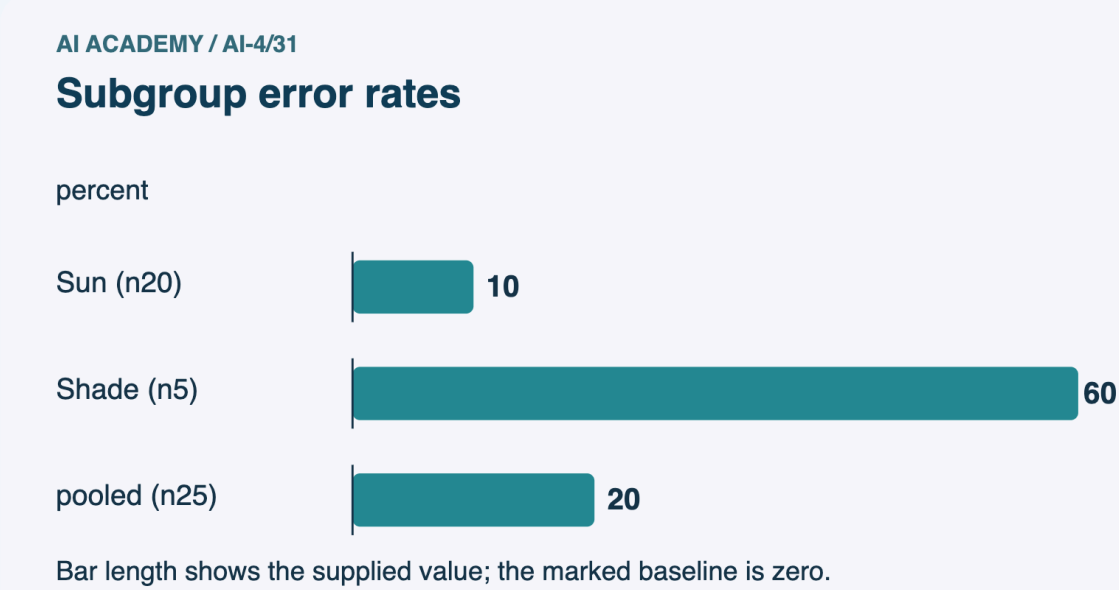


Follow the labelled picture with the matching explanation. Open the vector diagram

Sun rate: Sun error=2/20=10%. → Shade rate: Shade error=3/5=**60%**. → Pooled rate: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Subgroup error rates

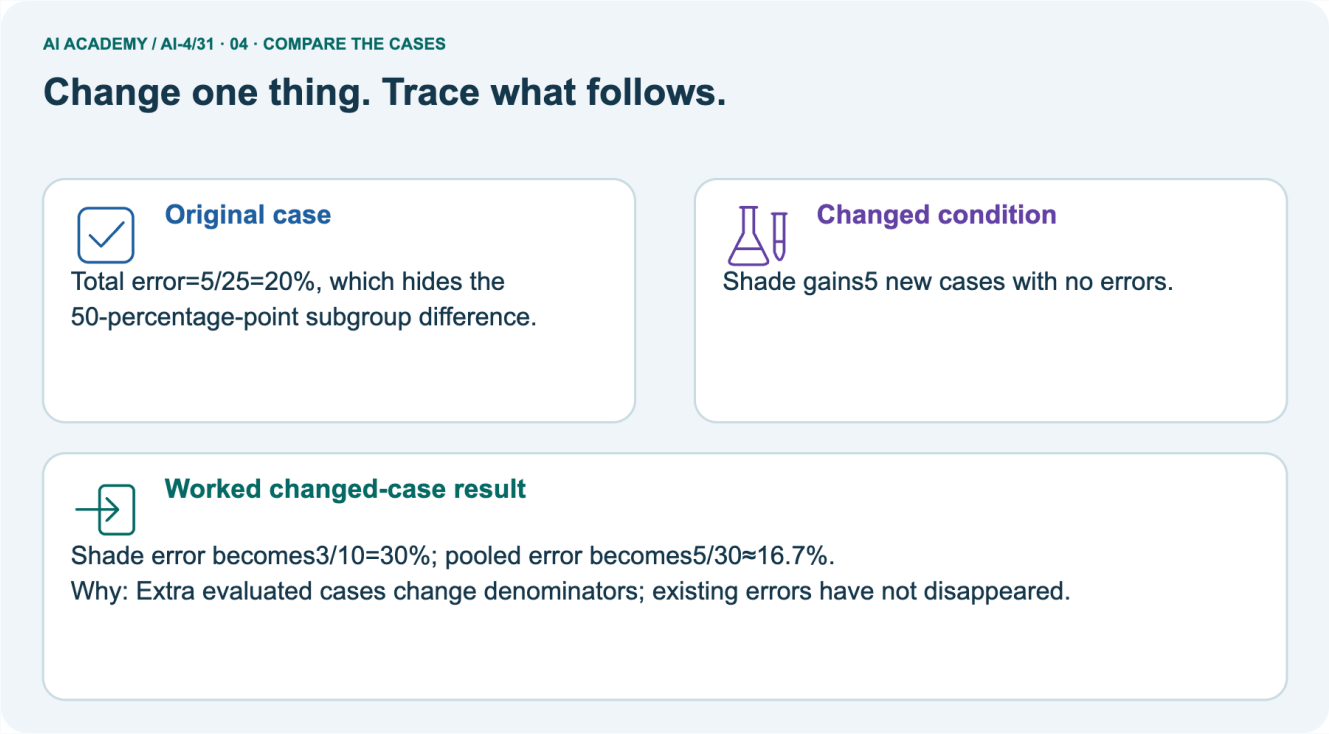


Follow the labelled picture with the matching explanation. Open the vector diagram

Subgroup error rates. Units: percent. Sun (n20): 10. Shade (n5): 60. pooled (n25): 20.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



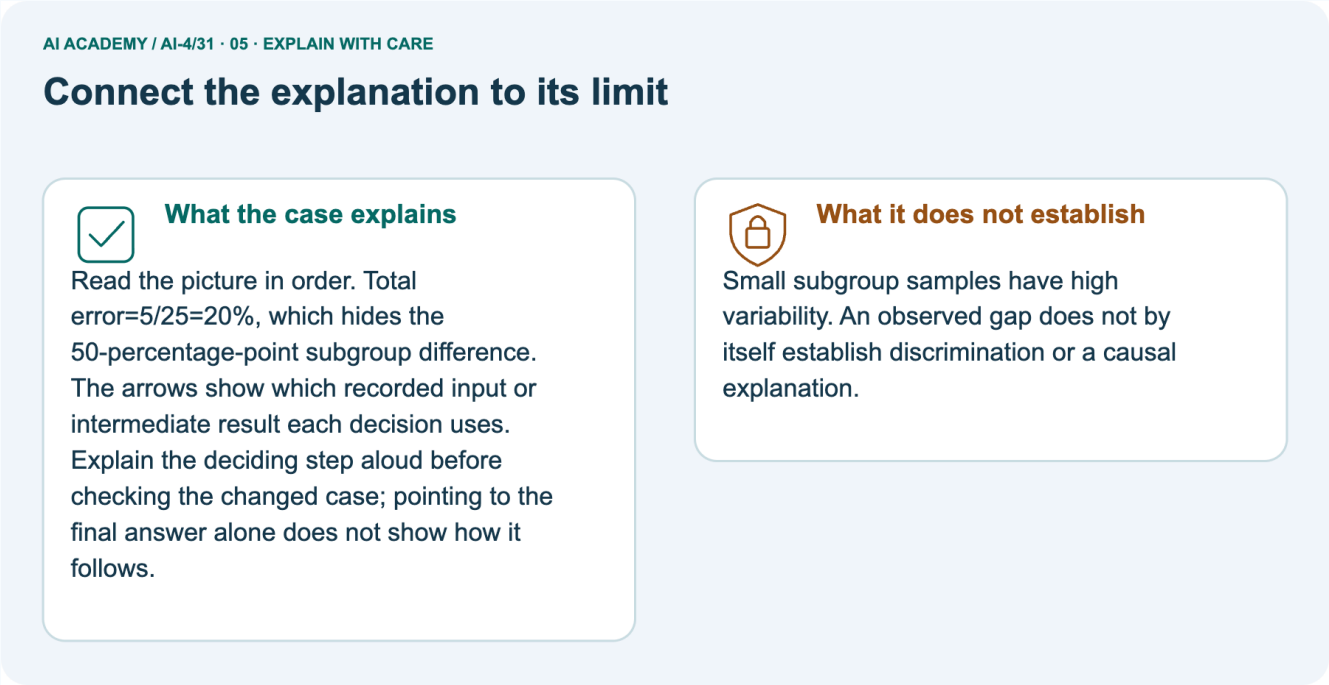
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Changed condition: Shade gains5 new cases with no errors.

Expected result: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

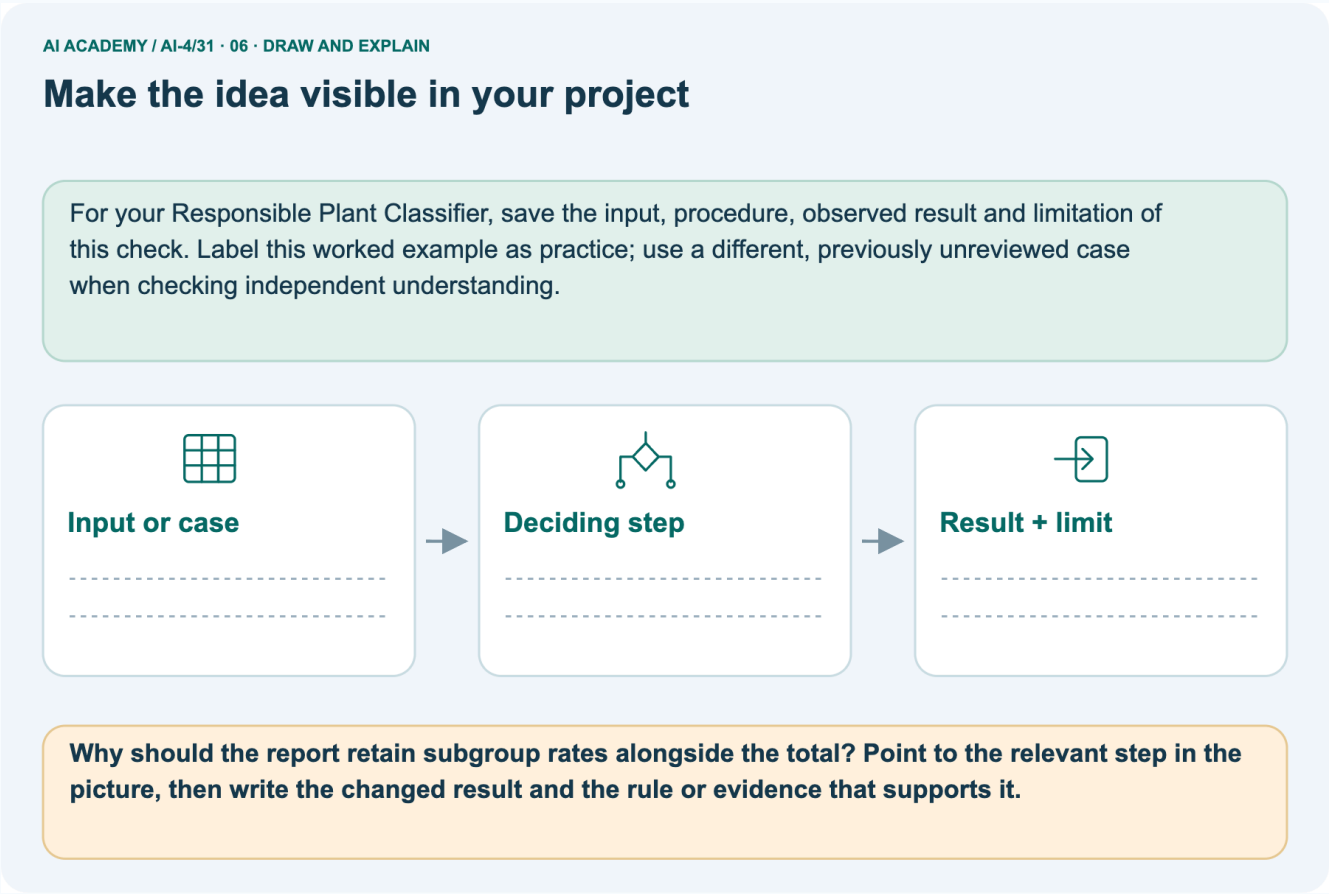
Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Small **subgroup** samples have high variability.

An observed gap does not by itself establish discrimination or a causal explanation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.